PH.45320 **FALL 2026**

Static Cosmology

Lecture 15: Final Project

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Overview

The final project merges one sky-cell update and reports its effect on the global catalog fit.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$CMT_{new} = merge(CMT_9, \Delta\theta_{cell})$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$L_{allcells} < L_{registry}$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$release = hash(manifest, map, catalog, log)$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Final Sky Cell Protocol
- Atlas Release Checklist
- Research Tool Attribution Guide